

GAME NO. 630
JANUARY, 1978

MIDWAY'S CLOWNS PARTS CATALOG



MIDWAY MFG. CO.

A BALLY COMPANY

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WARNING

**THIS GAME MUST BE GROUNDED. FAILURE TO DO
SO MAY RESULT IN DESTRUCTION TO ELECTRONIC
COMPONENTS.**

GENERAL INSTRUCTIONS FOR "CLOWNS"

INSTALLATION:

1. Remove shipping cleats located on the bottom of the cabinet. .
2. Install 4 provided leg levelers to bottom of cabinet and level cabinet.
3. The power is controlled by a switch located on top of the cabinet. Plug into A. C. only, 115 volts, 60 cycles. In low line areas (105 volts or less) a boost in the output voltage of the transformer may be obtained by removing the wire from the 115 volt tap of the transformer and rewiring to the 105 volt position.

VOLTAGE CONTROL POT:

Located on Power Supply (P.C. 0080-00904F) and is preset at the factory. These pots should not be tampered with unless the Distributor Service Department is contacted.

METHOD OF PLAY:

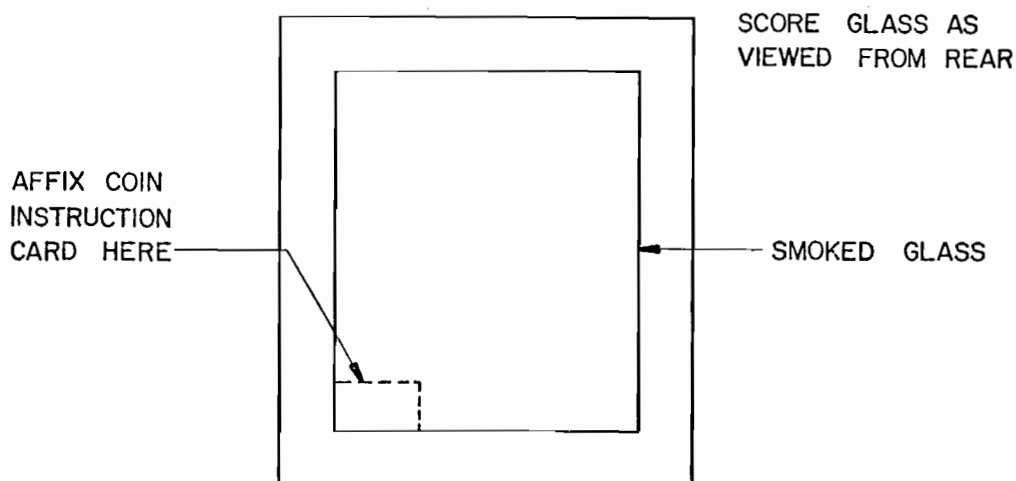
Coin, number of jumps and extra jump adjustment switch located on Game Logic Board (P.C. A 080-90700-B630) and may be adjusted as indicated on separate instruction sheet in back box area.

VARIABLE COIN SETTINGS:

This unit is shipped with the coin adjustment switches (#1 & #2 in "on" position) set in the one coin per play positions. When another coin option is desired, (see separate instructions in back box area) it is necessary to insert an additional coin instruction card.

TO INSERT NEW COIN INSTRUCTION CARD:

1. Remove Display Glass.
2. With the score glass lying face down, lift left corner of smoked glass slightly.
3. Insert New coin instruction card and tape to inside surface of score glass with clear tape.
4. Press smoked glass firmly back into place.



General Instructions For "Clowns" (continued)

TO SERVICE CONTROL PANEL POTS:

1. Remove Control Panel Cable jack from Game Logic Board.
2. Open front coin door and release (2) clamps located on each side of cabinet wall thru door opening.
3. Remove wire nut between main cable and Control Panel cable thru front door opening.
4. Remove control Panel and service same.

TO ADJUST SPRINGBOARD TRAVEL:

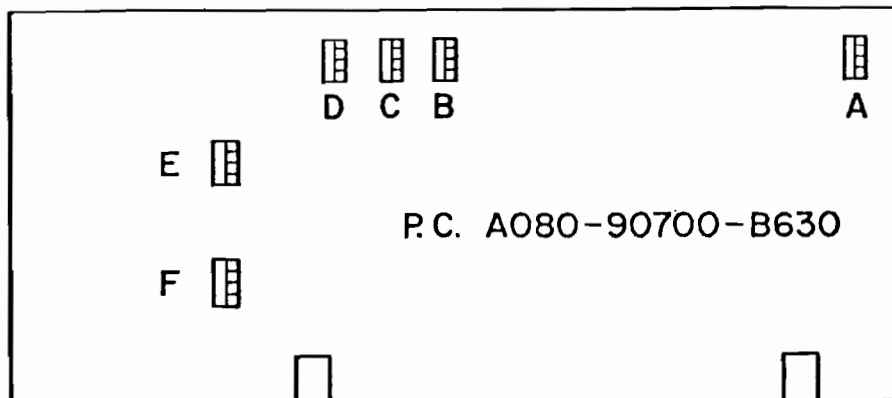
1. Place springboard alignment switch #7 in "OFF" position.
2. Rotate pots, A, B, C and D located on Game Logic Board (P.C. A080-90700-B630) to center position as indicated by arrow on pot faces.
3. Adjust each players control pot (located on Control Panel) to travel from approximately 1K ohms (lower limit) to 9.5K ohms. (upper limit).
4. Rotate player one control pot to left extremity. If springboard does not travel to left edge of screen, trim the travel by rotating pot B (500 ohm) until it does. If springboard over-travels left edge of screen rotate pot B in opposite direction.
5. Rotate Player One control pot to right edge of screen. If springboard does not travel to right edge of screen, trim the travel by rotating pot A (5K ohm) until it does. If springboard over travels right edge of screen rotate pot A in opposite direction.
6. Repeat steps 3 and 4 using player two control pot. Pot C (5K ohm) controls left edge trim. Pot D (500 ohm) control right edge trim.
7. Reset springboard alignment switch #7 to "ON" position.

VOLUME CONTROL POT:

Located on Game Logic Board (PC A080-90700-B630) raises or lowers all sounds and may be adjusted by rotating pot E as desired.

TONE CONTROL POT:

Located on Game Logic Board (PC A080-90700-B630) varies tone of music as desired by rotating pot F.



OLD METHOD OF PLAY ADJUSTMENTS

	SW. 1	SW. 2
1 COIN = 1 PLAYER	ON	ON
2 COINS = 2 PLAYERS		
1 COIN = 1 PLAYER	OFF	ON
1 COIN = 2 PLAYERS		
2 COINS = 1 PLAYER	ON	OFF
2 COINS = 2 PLAYERS		
2 COINS = 1 PLAYER	OFF	OFF
4 COINS = 2 PLAYERS		

NUMBER OF JUMP ADJUSTMENTS

	SW. 3	SW. 4
2 JUMPS PER GAME	ON	ON
3 JUMPS PER GAME	OFF	ON
4 JUMPS PER GAME	ON	OFF
5 JUMPS PER GAME	OFF	OFF

BALLOON RESET ADJUSTMENTS

	SW. 5
BALLOON RESET AFTER KNOCKING OUT EACH ROW	ON
BALLOON RESET AFTER KNOCKING OUT ALL BALLOONS	OFF

BONUS JUMP ADJUSTMENTS

	SW. 6
1 BONUS JUMP AWARDED AT 3000 SCORE	ON
1 BONUS JUMP AWARDED AT 4000 SCORE	OFF

SPRINGBOARD ALIGNMENT SWITCH

	SW. 7
PLAY MODE	ON
SPRINGBOARD ALIGNMENT	OFF

TEST

	SW. 8
PLAY MODE	ON
RAM-ROM TEST	OFF

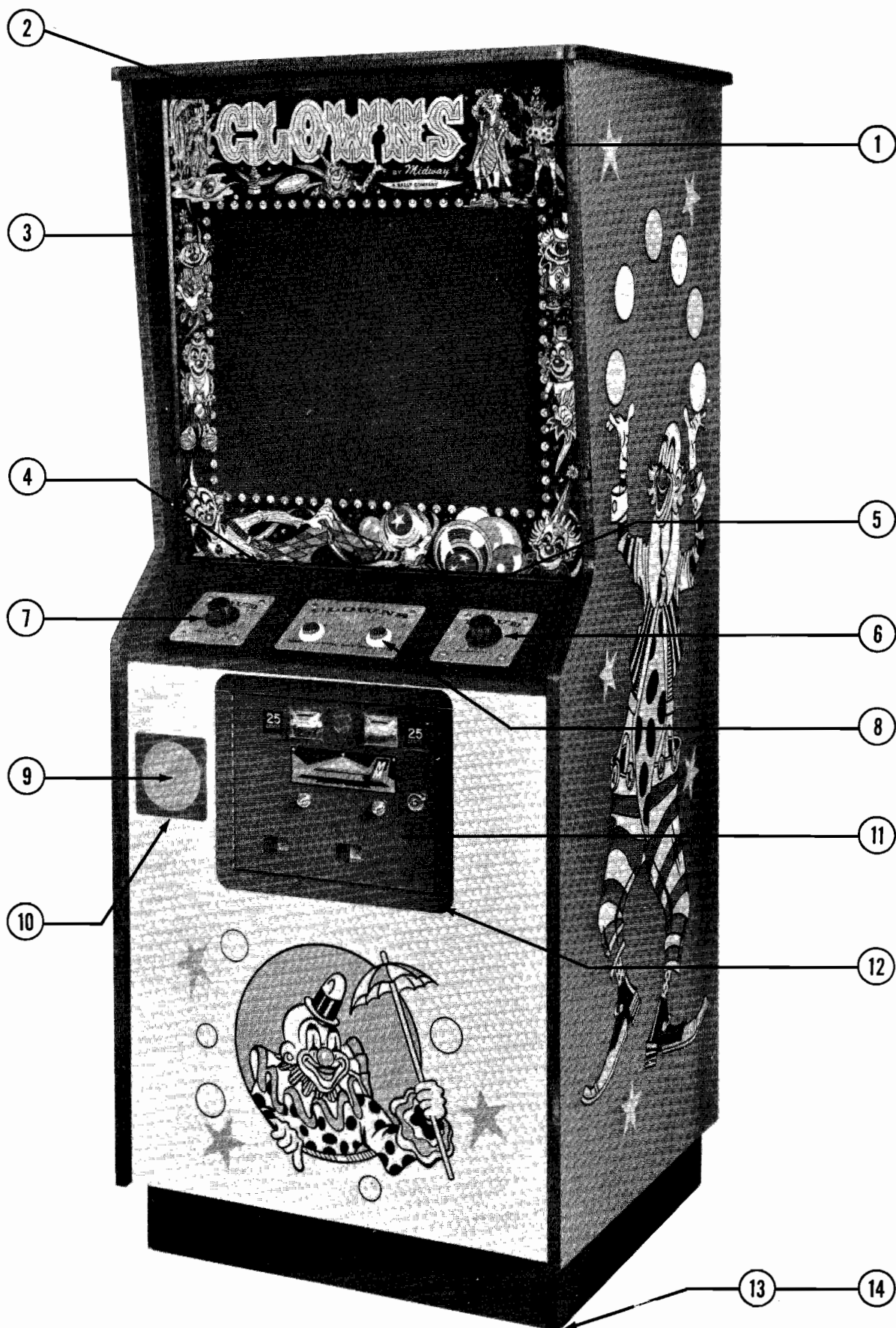
0630-CLNS

NEW METHOD OF PLAY ADJUSTMENTS

	SW. 1	SW. 2
1 COIN = 1 PLAYER	ON	ON
2 COINS = 2 PLAYERS		
1 COIN = 1 PLAYER	OFF	ON
1 COIN = 2 PLAYERS		
2 COINS = 1 PLAYER	ON	OFF
2 COINS = 2 PLAYERS		
2 COINS = 1 PLAYER	OFF	OFF
4 COINS = 2 PLAYERS		
	SW. 3	SW. 4
NO BONUS GAME	ON	ON
BONUS GAME AT 9,000	OFF	ON
11,000	ON	OFF
13,000	OFF	OFF
		SW. 5
BALLOONS RESET AFTER KNOCKING OUT EACH ROW		ON
BALLOONS RESET AFTER KNOCKING OUT ALL BALLOONS		OFF
		SW. 6
1 BONUS JUMP AWARDED AT 3,000 SCORE		ON
1 BONUS JUMP AWARDED AT 4,000 SCORE		OFF
		SW. 7
3 JUMPS PER GAME		ON
4 JUMPS PER GAME		OFF
		SW. 8
PLAY MODE		ON
OPEN COIN SW FOR RAM/ROM		OFF
CLOSE COIN SW FOR ALIGNMENT SPRING BOARD		

*Service Bulletin will advise of ROM Changes due to new method of play adjustments and serial no's. of games with new and old method of play adjustments.

NO. 630 – CLOWNS – PICTURE



NO. 630 – CLOWNS – PICTURE

ORDER BY PART NUMBER ONLY

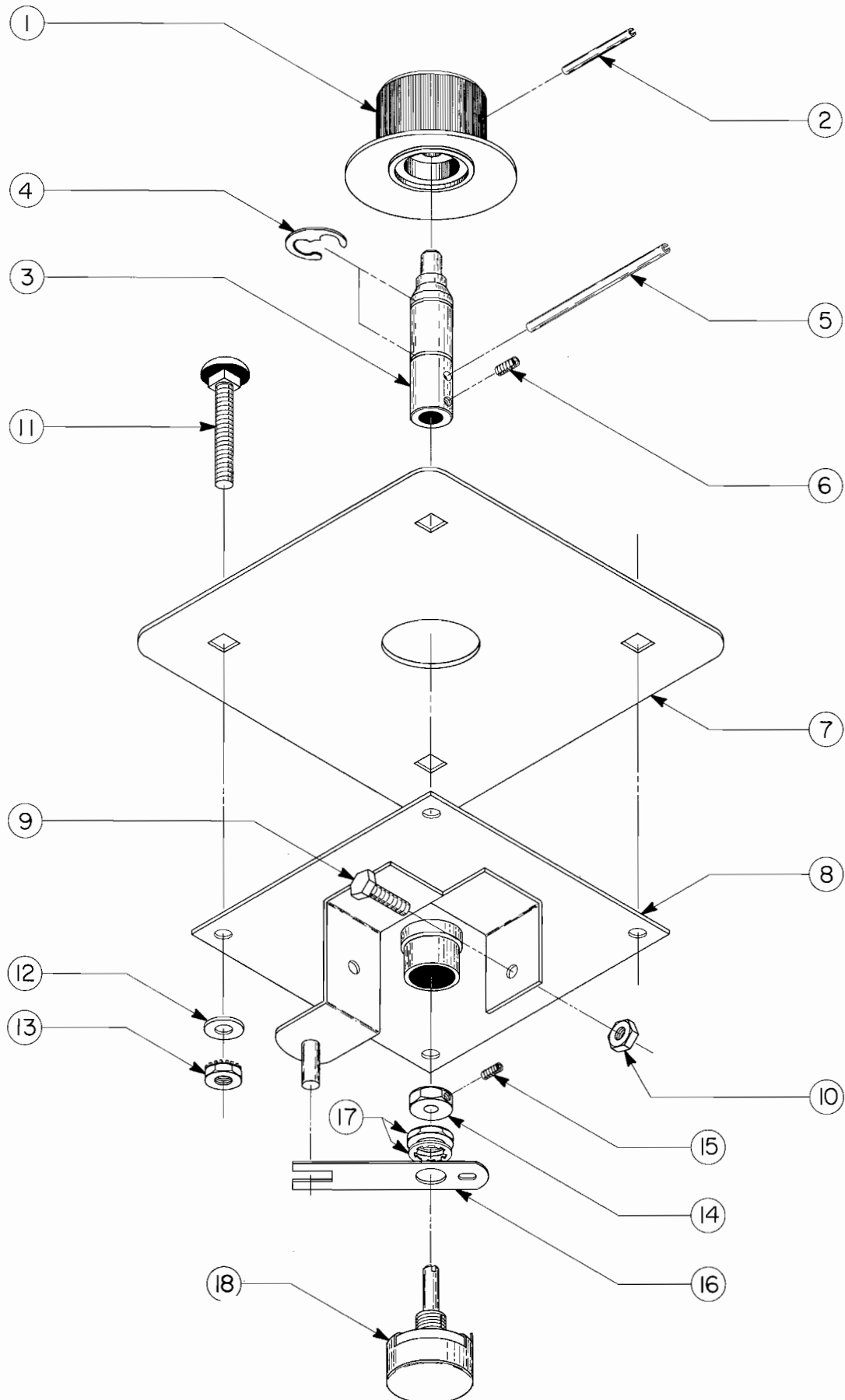
ITEM	PART NO	DESCRIPTION
1	0630-00900-01SC	DISPLAY PLEX
2	0537-00903-3100	CYCOLAC EDGEING
3	0537-00903-3200	CYCOLAC EDGEING (2 REQ'D.)
4	0537-00138-0000	GLASS LIFT CHANNEL
5	A630-00007-0000	CONTROL PANEL ASSY. (INCLUDED ARE ITEMS 6 THRU 8)
6	A630-00012-0000	PLAYER #1 CONTROL ASSY. – RIGHT
7	A630-00013-0000	PLAYER #2 CONTROL ASSY. – LEFT (SAME ASSY. AS THE RIGHT)
8	A630-00014-0000	PUSH BUTTON ASSY. (2 REQ'D.)
	0630-00906-0000	BUTTON PLATE
	0586-00036-0000	PUSH BUTTON SWITCH ASSY. (2 REQ'D.)
	0020-00202-0000	SWITCH PLATE (2 REQ'D.)
	0017-00101-0521	#5-40 x 3/8 R.H.M.S. (4 REQ'D.)
	0586-00107-0000	MOUNTING BRACKET (2 REQ'D.)
9	0549-00128-0000	SPEAKER GRILL
10	0569-00001-00GR	SPEAKER BEZEL ASSY.
	0017-00003-0161	SPEAKER
	0017-00041-0576	SPEAKER GASKET
	0017-00104-0022	#6 STL. WASHERS (4 REQ'D.)
	0017-00103-0005	#6-32 HEX NUTS (4 REQ'D.)
11	A090-00020-02GR	DOUBLE ENTRY COIN DOOR ASSY.
12	0090-00002-02GR	COIN DOOR FRAME
	0017-00101-0555	#6-32 x 5/16 HEX HD. BLK. S.M.S. (3 REQ'D.) (MOUNTS COIN DOOR TO FRAME)
13	0017-00102-0048	LEG LEVELERS (4 REQ'D.)
14	0017-00103-0026	LEG LEVELER NUTS (4 REQ'D.)
		<u>NOT SHOWN LIST</u>
	0017-00003-0189	23" MONITOR (CHASSIS & TUBE COMPLETE)
	0017-00101-0639	#8-32 x 1-1/4 CARRIAGE BOLTS (4 REQ'D.)
	0017-00104-0022	#8 STEEL WASHER (4 REQ'D.)
	0017-00103-0061	#8-32 HEX LOCK NUTS S.E.M.S. (4 REQ'D.)
	0080-00101-0000	FISHPAPER SHIELD
	0017-00009-0033	BASSICK CLAMP (2 REQ'D.)
	0603-00131-0000	STRIKE (2 REQ'D.)
	0630-00901-0000	TUBE MASK

NO. 630 – CLOWNS – PICTURE

ORDER BY PART NUMBER ONLY

ITEM	PART N ^o	DESCRIPTION
	0605-00104-0100	GLASS LOCK BRKT. (2 REQ'D.)
	0017-00101-0028	#8 x 3/4 HEX. HD. BLK. WD. SCREW (4 REQ'D.)
	0630-00902-0000	GRAY PLEXIGLASS – 18" x 22" x 1/8" – #2074
	0630-00903-0000	TAPE – 1/2" x 19"
	0630-00904-0100	CHARTPACK TRANSPARENT TAPE – 3/4" x 20" BLUE
	0630-00904-0200	CHARTPACK TRANSPARENT TAPE – 3/4" x 20" GRN
	0630-00904-0300	CHARTPACK TRANSPARENT TAPE – 3/4" x 20" YEL
	0567-00106-0000	SWITCH PLATE
	0586-00036-0000	ON-OFF SWITCH
	0630-00504-0000	DISPLAY INSERT KIT – NOT ON ALL GAMES, BUT MAY BE PURCHASED & INSTALLED ON ANY CLOWN-GAME IF DESIRED.

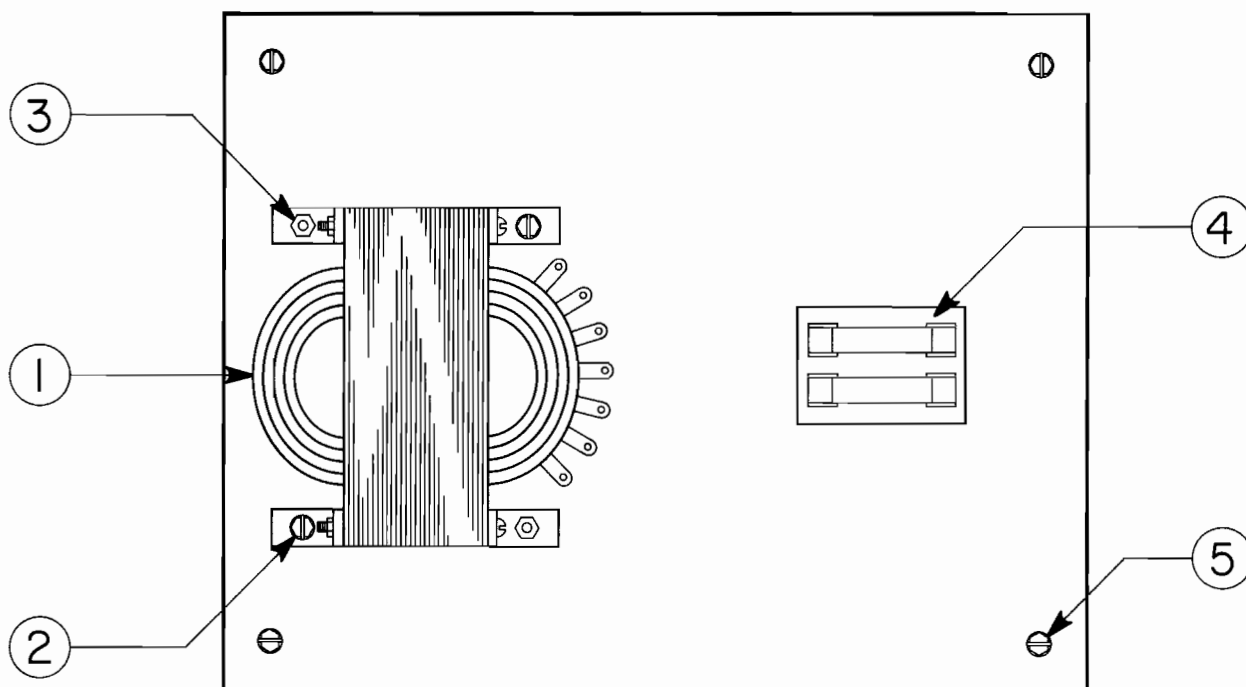
NO. 630 – CLOWNS – PLAYER CONTROL ASSY.



NO. 630 – CLOWNS – PLAYER CONTROL ASSY.
ORDER BY PART NUMBER ONLY

ITEM	PART N^o	DESCRIPTION
1	A630-00009-0000	KNOB ASSY.
2	0017-00007-0011	3/32 x 1-1/8 ROLL PIN
3	0630-00700-0000	SHAFT
4	0017-00100-0050	"E" RING (2 REQ'D.)
5	0017-00007-0083	1/8 x 1-5/8 ROLL PIN
6	0017-00101-0308	#8-32 x 3/16 SET SCREWS (2 REQ'D.)
7	0630-00905-0100	PLAYER #1 CONTROL PLATE
	0630-00905-0200	PLAYER #2 CONTROL PLATE (NOT SHOWN)
8	A630-00008-0000	MOUNTING PLATE ASSY.
9	0017-00101-0704	#10-32 x 1/2 HEX HD. NYLON M.S. (2 REQ'D.)
10	0017-00103-0010	#10-32 HEX HD. NUT (2 REQ'D.)
11	0017-00101-0639	#8-32 x 1-1/4 CARRIAGE BOLTS (4 REQ'D.)
12	0017-00104-0022	#8 STL. FLAT WASHER (4 REQ'D.)
13	0017-00103-0061	#8-32 HEX LOCKNUT (4 REQ'D.)
14	0567-00700-0000	ADJUSTING NUT
15	0017-00101-0302	#6-32 x 3/16 SET SCREW
16	0591-00105-0100	LOCATOR – POT
17		HARDWARE FOR ITEM #18
18	0063-27705-16KX	25K – POT
	A630-00012-0000	PLAYER #1 CONTROL ASSY. – COMPLETE
	A630-00013-0000	PLAYER #2 CONTROL ASSY. – COMPLETE

NO. 630 – CLOWNS – TRANSFORMER BOARD ASSY.



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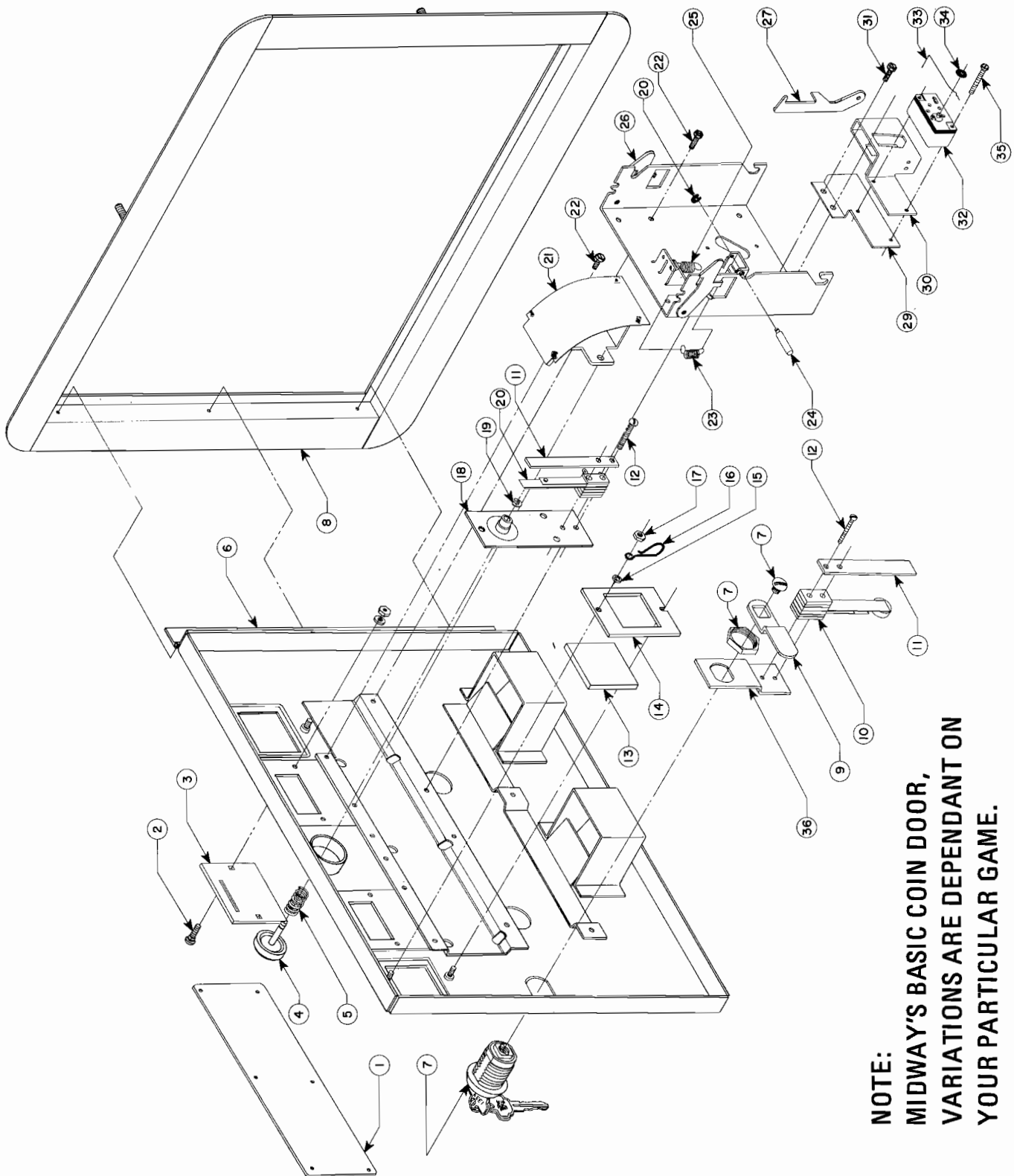
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NO. 630 – CLOWNS – TRANSFORMER BOARD ASSY.

ORDER BY PART NUMBER ONLY

ITEM	PART N ^o	DESCRIPTION
1	MT00-00058-A000 MT00-00059-A000	TRANSFORMER – 60 CYCLE - OR - TRANSFORMER – 50 CYCLE – (FOREIGN)
2	0017-00101-0025	#8 x 1/2 SLOT. HEX HD. S.M.S. (2 REQ'D.)
3	0017-00101-0692 0017-00103-0008	#8-32 x 7/8 CARRIAGE BOLTS (2 REQ'D.), WITH #8-32 HEX NUT (2 REQ'D.)
4	0720-00001-0200	FUSE BLOCK ASSY. (2 POSITION)
5	0017-00101-0030 0017-00101-0017	#8 x 1" HEX HD. WD. SCREW (4 REQ'D.) #6 x 1/2 HEX HD. S.M.S. (4 REQ'D.) - <u>NOT SHOWN</u>
	0017-00003-0064 A630-00015-0000	LINE CORD – NOT SHOWN TRANSFORMER BOARD ASSY. – COMPLETE

NO. 630 – CLOWNS – FRONT DOOR ASSY.



NOTE:
MIDWAY'S BASIC COIN DOOR,
VARIATIONS ARE DEPENDANT ON
YOUR PARTICULAR GAME.

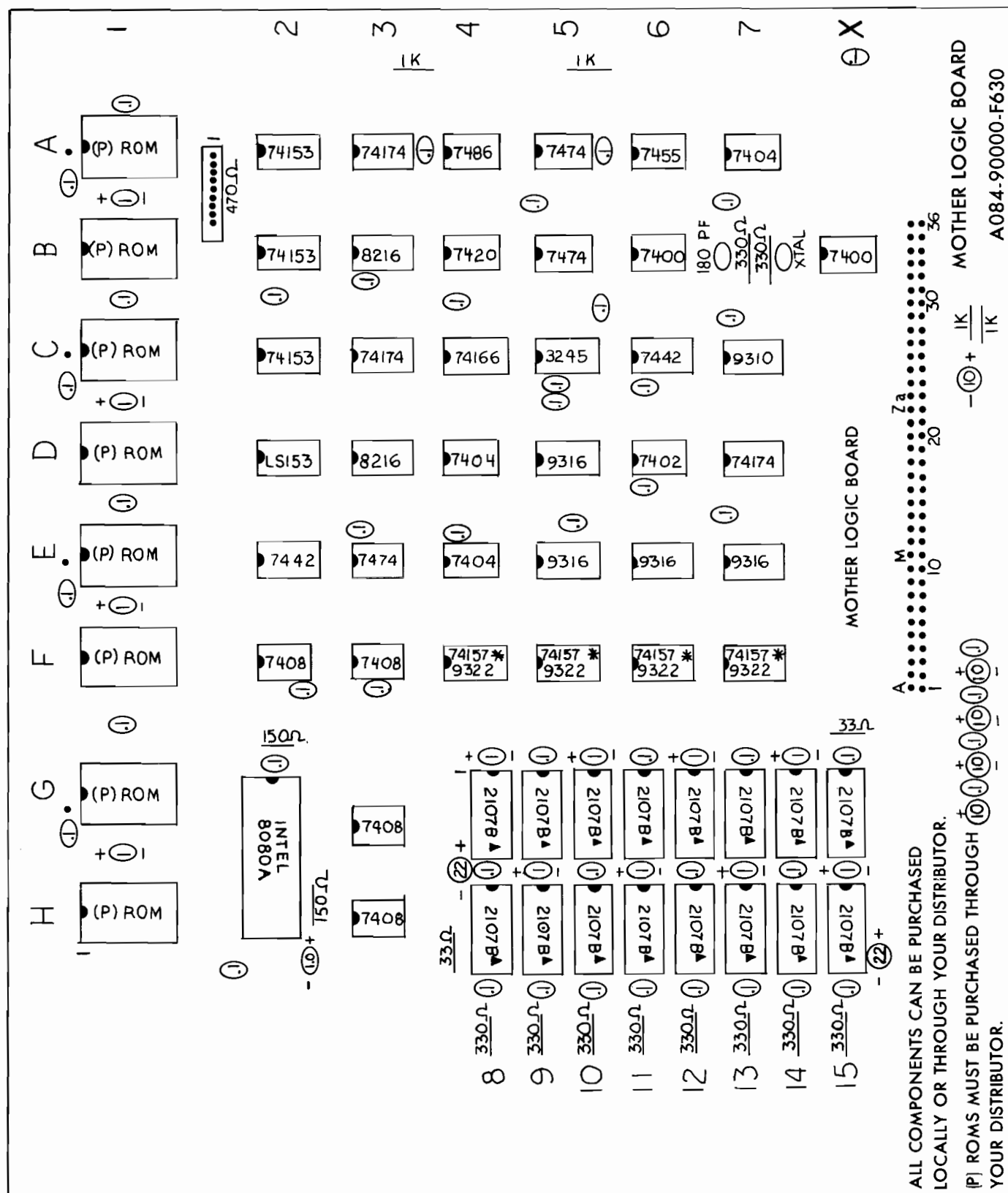
NO. 630 – CLOWNS – FRONT DOOR ASSY.

ORDER BY PART NUMBER ONLY

ITEM	PART N^o	DESCRIPTION
1	0090-00902-0000 0090-00902-0100	DRESS PLATE DRESS PLATE (W/O PRESS BUTTON TO START GAME)
2	0017-00101-0552	6-32 x 1/4 CARRIAGE BOLT (2)
3	0090-00117-0003	COIN ENTRY PLATE (25¢)
4	0090-00906-0000	PLASTIC START BUTTON
5	0010-00007-0000	COMPRESSION SPRING
6	0090-00013-0200	DOOR ASSY. DOUBLE ENTRY
7	0017-00005-0050	DOOR LOCK & KEY WITH SCREW & NUT
8	0090-00002-02BR	DOUBLE DOOR FRAME
9	0017-00005-0041	421 N.S. CAM
10	0090-00901-0000	DOOR SWITCH
11	0090-00126-012N	SWITCH BACKUP PLATE
12	0017-00101-0528	5-40 x 3/4" LNG. M.S.
13	0090-00903-0040	25¢ WINDOW
14	0090-00143-0000	COIN PLEX RETAINER
15	0017-00104-0002	SPLIT LOCK WASHER
16	0017-00007-0019	KEY HOOK
17	0017-00103-0005	6-32 HEX NUT
18	0090-00010-0000	SW. PLATE & OILLITE ASSY.
19	0017-00100-0018	"E" RING
20	0090-00131-0000	SWITCH
21	0090-00104-0000	TOP & BTM. COIN CHUTE W/BRKT. ASSY.
22	0017-00101-0598	8-32 x 5/16 SCREW
23	0010-00181-0100	SPRING
24	0090-00129-0000	PIVOT POST
25	0010-00134-0000	SPRING
26	0090-00008-0000	ACCEPTOR FRAME ASSY.
27	0093-00155-0000	REJECTOR LEVER
28	0017-00100-0012	"E" RING
29	0093-00151-0000	COIN GUIDE MTG. BRACKET
30	0090-00005-0000	COIN GUIDE ASSY.
31	0017-00101-0555	6-32 x 5/16 SCREW
32	0017-00005-0039	COIN SWITCH
33	0010-00557-0025	TRIP WIRE
34	0017-00007-0132	PUSH-ON RING (BIK)
35	0017-00101-0507	#4-36 x 3/4 SCREW
36	0090-00128-0000	SWITCH BRKT. – DOOR TILT

NO. 630 – CLOWNS – ADDITIONAL PARTS LIST
ORDER BY PART NUMBER ONLY

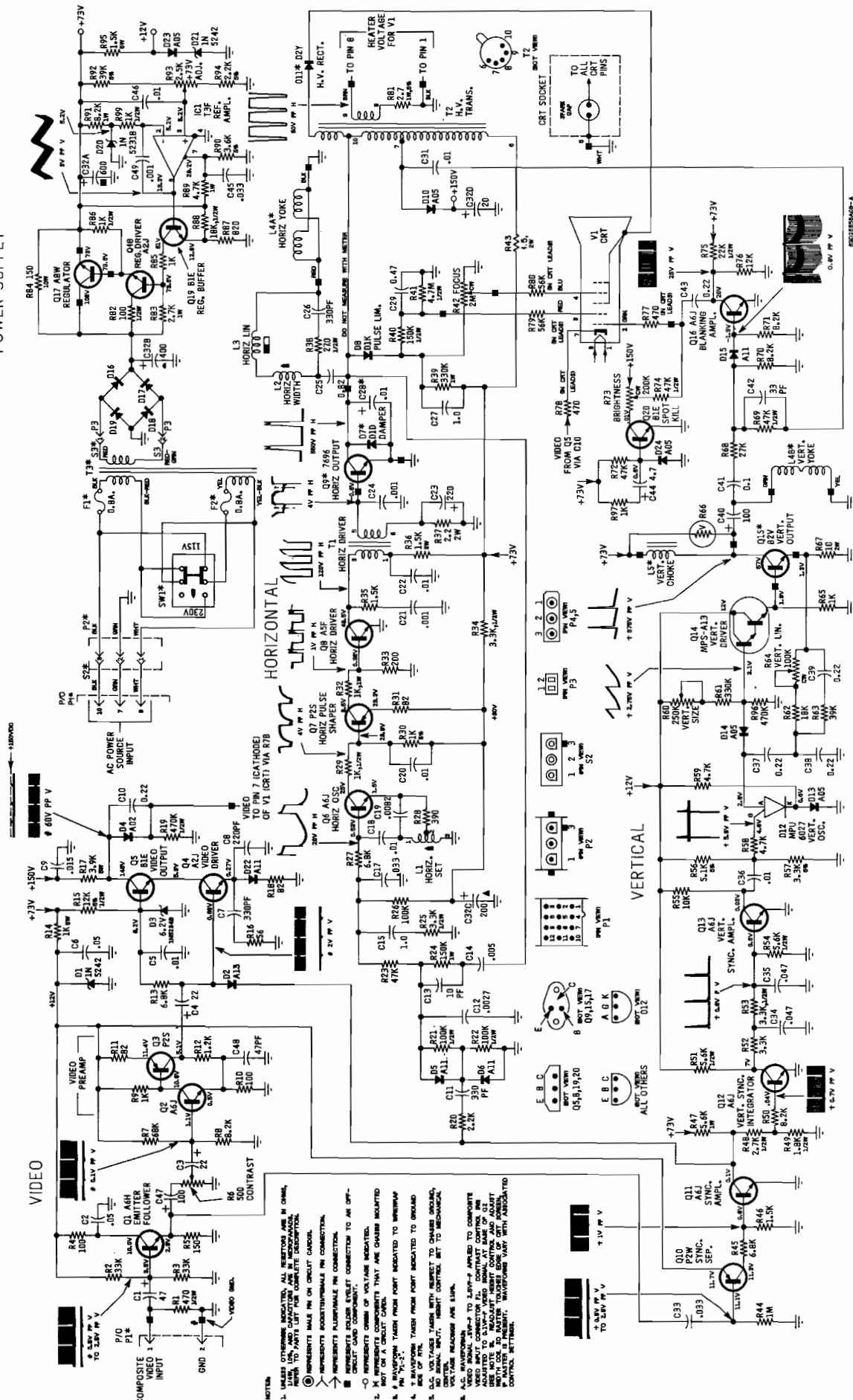
ITEM	PART N^o	DESCRIPTION
	A630-00003-0000	UPPER BACK DOOR ASSY.
	A630-00002-0000	LOWER BACK DOOR ASSY.
	A623-00110-0000	COIN BOX ASSY.
	A623-00013-0000	COIN BOX COVER ASSY.
	A624-00001-0000	COIN BOX GUIDE BRKT. ASSY.
	0624-00101-0000	COIN BOX GUIDE BRKT.
	0017-00101-0628	#8-32 x 3/4 CARRIAGE BOLTS (4 REQ'D.)
	0017-00104-0022	#8 STEEL WASHER (4 REQ'D.)
	0017-00103-0008	#8-32 HEX NUT (4 REQ'D.)
	0017-00003-0102	COIN METER (MOUNTED ON COIN DOOR)
	0017-00009-0009	TURN BUTTONS (2 REQ'D.)
	0017-00101-0028	#8 x 3/4 HEX HD. WD. SCREWS (2 REQ'D.)
	0582-00100-0000	VENT SCREENS (2 REQ'D.)
	A630-00014-0000	PUSH BUTTON ASSY. – (MAKING UP THE NEXT 6 PART NUMBERS.)
	0630-00907-0000	BUTTON (0017-00009-0384) DYED RED
	0017-00009-0376	HOUSING
	0010-00242-0000	SPRING
	0017-00104-0028	FLAT WASHER
	0017-00100-0025	"E" RING
	0017-00103-0054	PAL NUT
	A630-00006-0000	MOTHER & GAME BOARD ASSY.
	A084-90000-F630	MOTHER BOARD ASSY.
	A084-90700-B630	GAME BOARD ASSY.
	A630-00015-0000	TRANSFORMER BOARD ASSY.
	A082-90400-E000	POWER SUPPLY BOARD ASSY.
	0597-00012-0100	P.C. SUPPORT BRKT. ASSY. – RIGHT HAND
	0597-00012-0200	P.C. SUPPORT BRKT. ASSY. – LEFT HAND
	0624-00902-0100	P.C. SUPPORT BRKT. – 12" (4 REQ'D.)
	0624-00902-0300	P.C. SUPPORT BRKT. – 2-1/2 (4 REQ'D.)
	0017-00101-0025	#8 x 1/2 HEX HD. WD. SCREWS (16 REQ'D.) FOR ALL P.C. SUPPORT BRKTS.)
	0017-00009-0419	PIN EXTRACTOR TOOL (PIN POPPER)



ALL COMPONENTS CAN BE PURCHASED
LOCALLY OR THROUGH YOUR DISTRIBUTOR.
[P] ROMS MUST BE PURCHASED THROUGH
YOUR DISTRIBUTOR.



POWER SUPPLY



M5000-155, M7000-155 - Diagram Schematic